Independent University, Bangladesh

Department of Computer Science & Engineering

CSE303L: Database Management System Lab

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Instructor: Noureen Islam

GROUP NO.: 15

TOPIC: Smart Healthcare Management System

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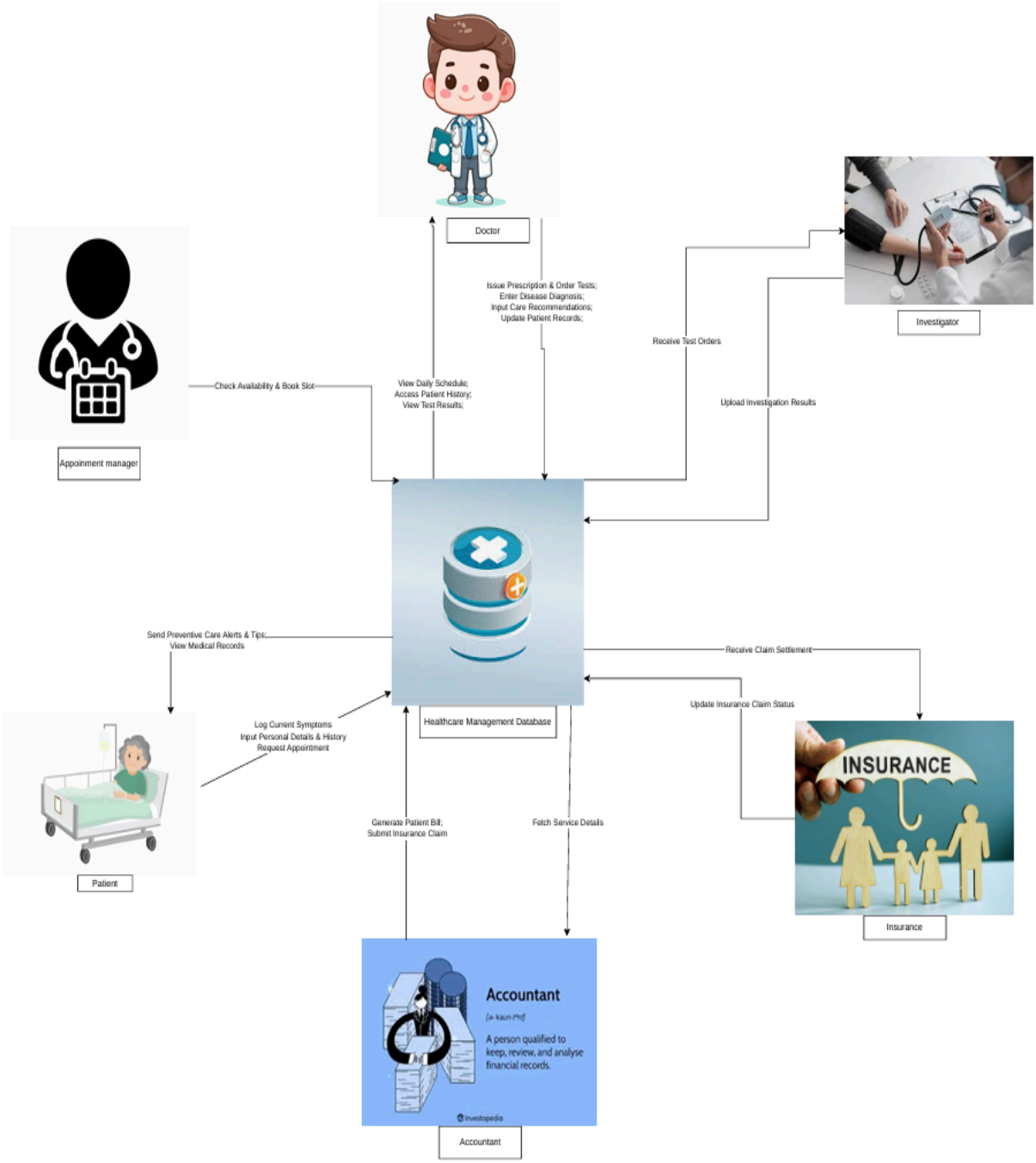
TABLE OF CONTENTS

TOPIC: [Smart Healthcare Management System] **2**
RICH PICTURE **2**
1.1 Rich Picture 2
1.2 Explanation 3
USER INTERFACE **4**
2.1 Webpage(s) for FEATURE 1 4
2.2 Webpage(s) for FEATURE 2 5
2.3 Webpage(s) for FEATURE 3 6
2.4 Webpage(s) for FEATURE 4 9
2.5 Webpage(s) for FEATURE 5 11
2.6 Webpage(s) for FEATURE 6 12
ENTITY RELATIONSHIP DIAGRAM (ERD) **13**
3.1 Entity Relationship Diagram (ERD) 13
3.2 Explanation 13
DATABASE WITH TABLE (populate the table(s) with data) **16**
4.1 Table for FEATURE 1 16
4.2 Table for FEATURE 2 17
4.3 Table for FEATURE 3 17
4.4 Table for FEATURE 4 18
4.5 Table for FEATURE 5 18
4.6 Table for FEATURE 6 19
GITHUB REPOSITORY & PROJECT DEMONSTRATION VIDEO **19**
5.1 Github Repository 19
5.2 Project Demonstration Video 19

TOPIC: [Smart Healthcare Management System]

RICH PICTURE

1.1 Rich Picture



1.2 Explanation

1. Centralized Data Core

At the heart of the system is the **Healthcare Management Database**. This acts as the single source of truth, ensuring that every actor—from the Doctor to the Accountant—accesses synchronized, real-time data. This eliminates physical paperwork and reduces communication delays between departments.

2. Clinical Workflow & Patient Care

- **The Patient Experience**: Patients initiate the process by logging current symptoms and requesting appointments. They also receive preventive care alerts and can view their own medical records directly from the database.
- **Medical Diagnosis**: **Doctors** interact with the core by viewing their daily schedules and accessing patient histories. They input disease diagnoses, issue prescriptions, and order specialized medical tests.
- **Laboratory Investigation**: The **Investigator** receives test orders from the database, conducts the necessary medical exams, and uploads the results back into the system for the Doctor's review.

3. Administrative & Financial Management

- **Operational Coordination**: The **Appointment Manager** serves as the system's gatekeeper, checking doctor availability in real-time and booking time slots to prevent over-scheduling.
- **Billing & Accounting**: Once medical services are rendered, the **Accountant** fetches service details from the database to generate patient bills and submit insurance claims.
- **Insurance Adjudication**: The **Insurance Officer** reviews the submitted claims. They update the claim status (Approved/Rejected) and coordinate the final claim settlement, ensuring the hospital's financial lifecycle is completed.

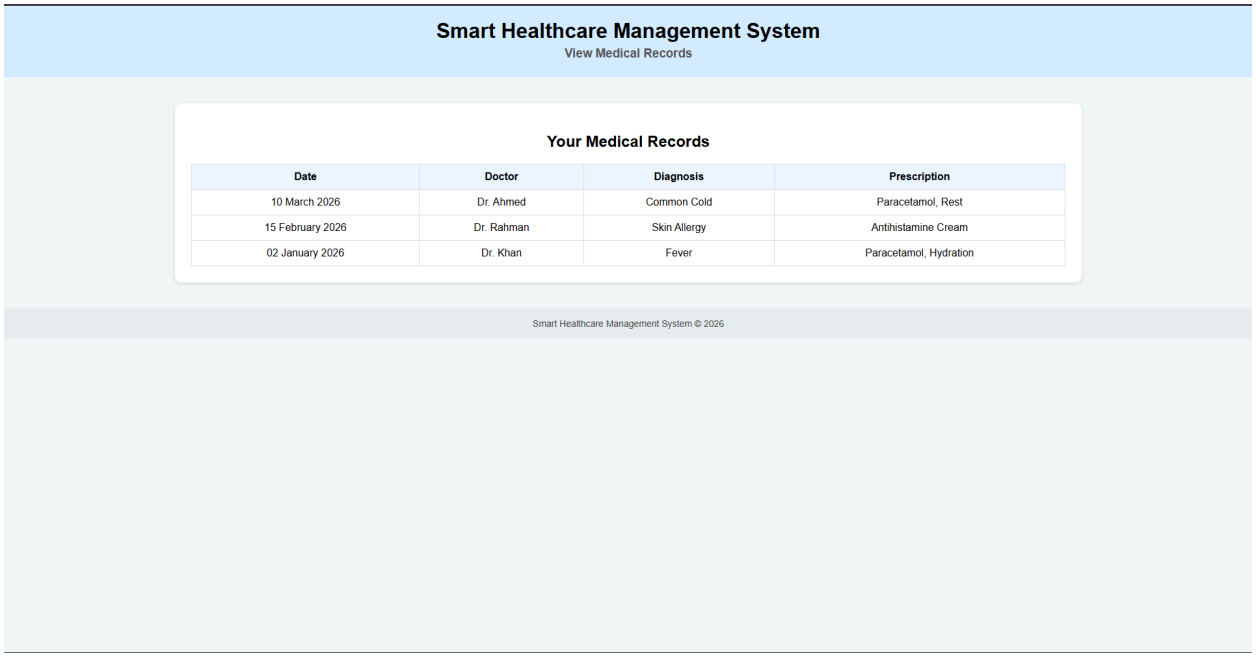
4. Systemic Benefits

- **Interdepartmental Connectivity**: The picture shows that no department works in a vacuum. For example, the Accountant cannot bill a patient until the Doctor or Investigator has updated the service details in the database.
- **Efficiency and Accuracy**: By automating these connections, the system minimizes human error in medical records and ensures that billing matches the actual treatments provided.

USER INTERFACE

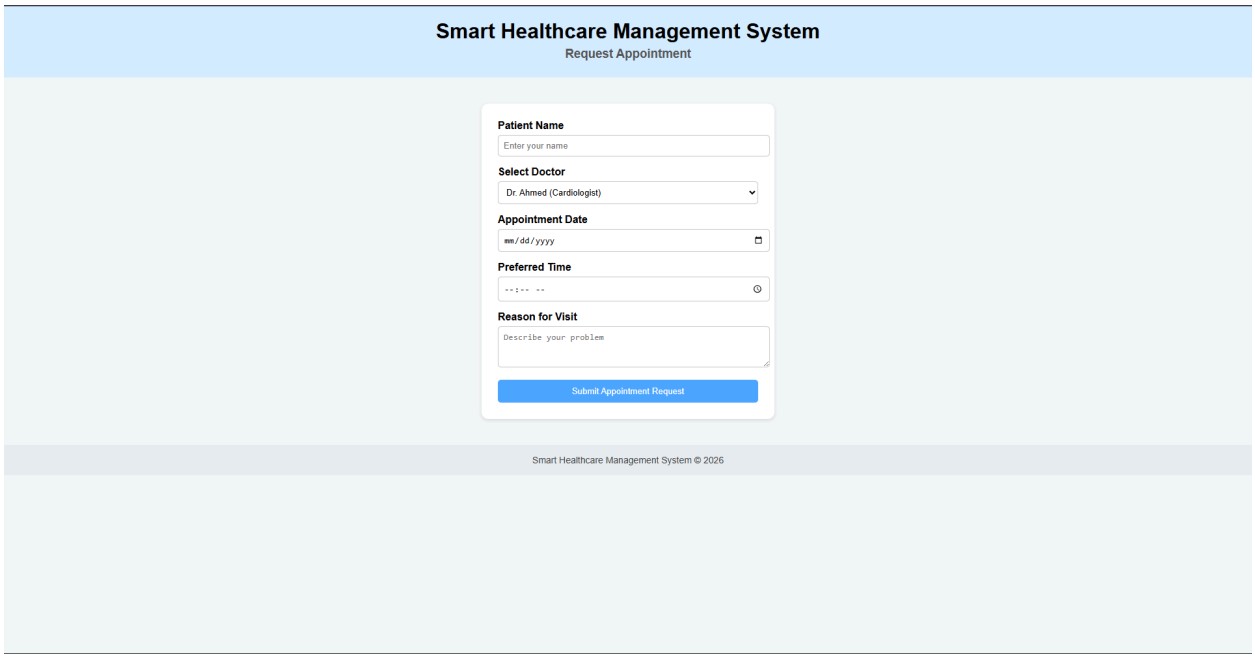
2.1 Webpage(s) for FEATURE 1

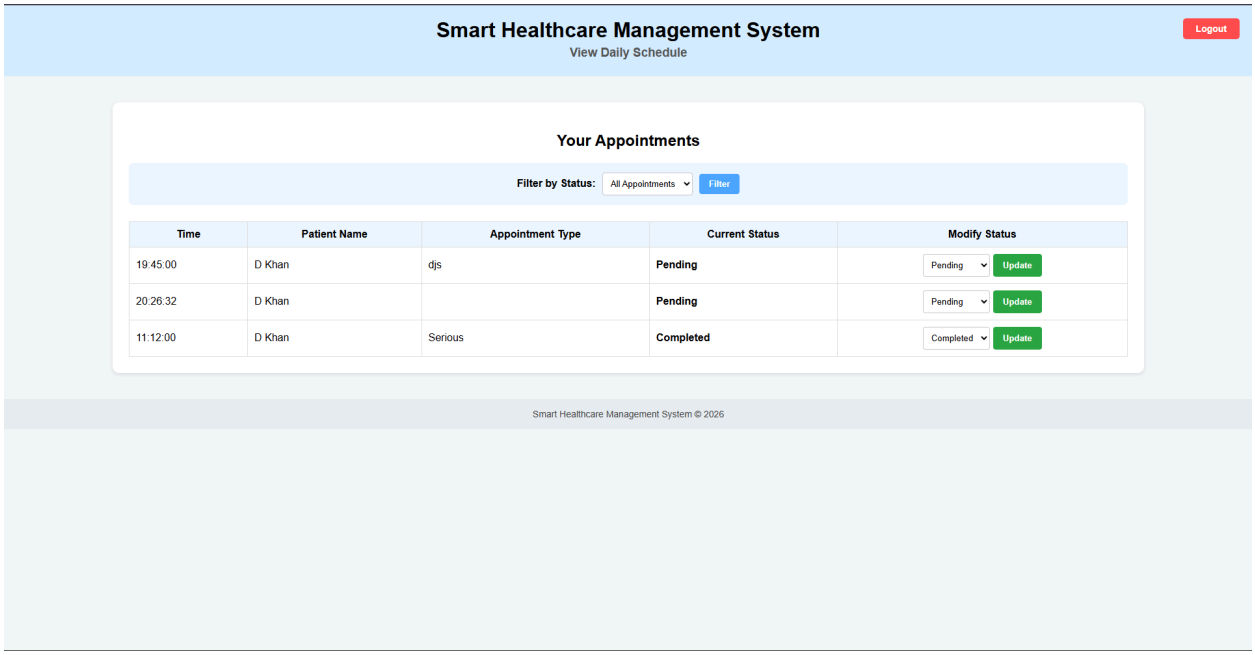
Webpage for FEATURE 1: Patient & Medical Record Management

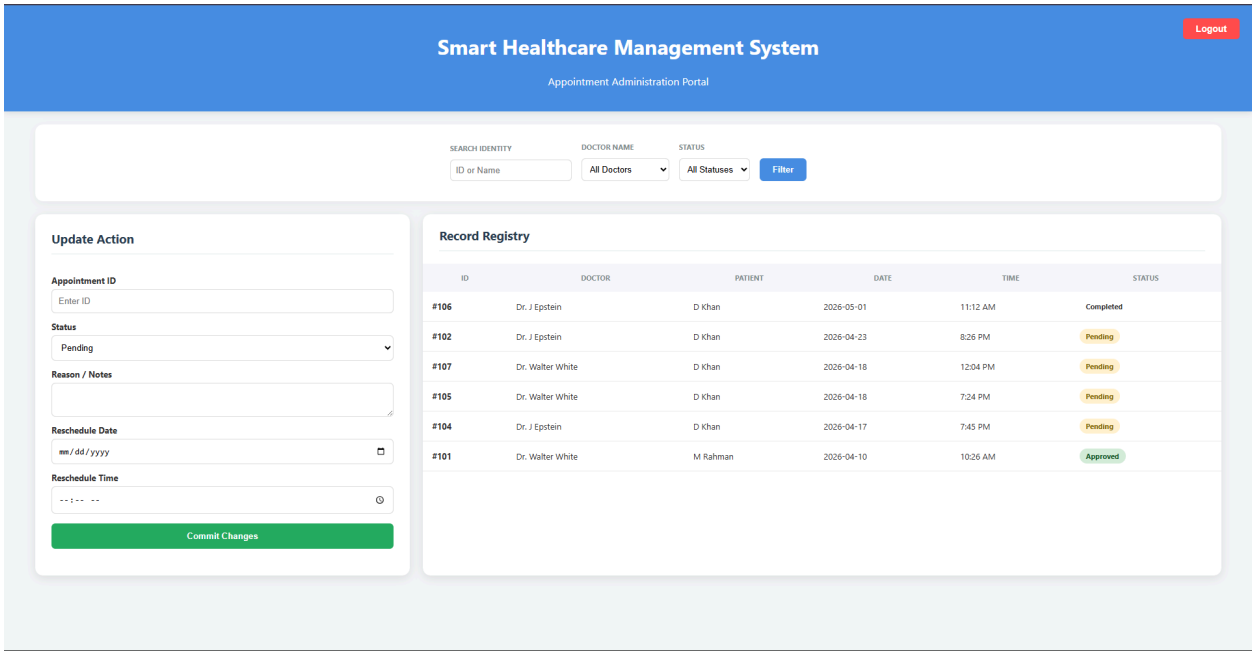


2.2 Webpage(s) for FEATURE 2

Webpage for FEATURE 2: Doctor & Appointment Management

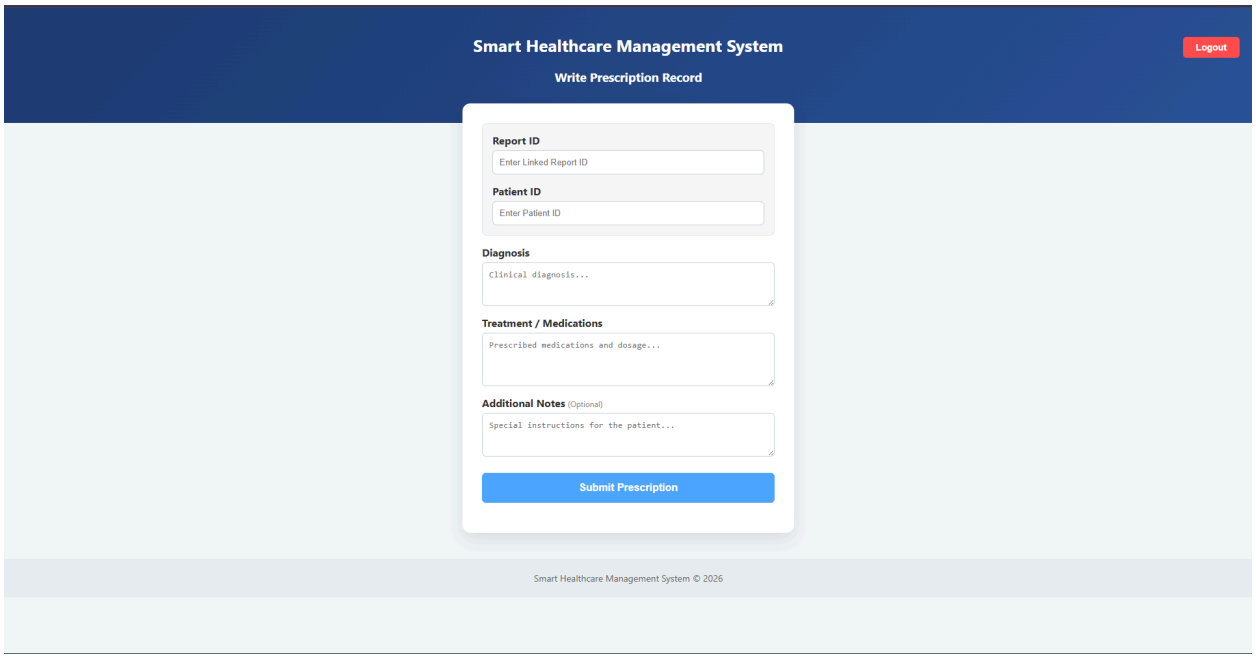


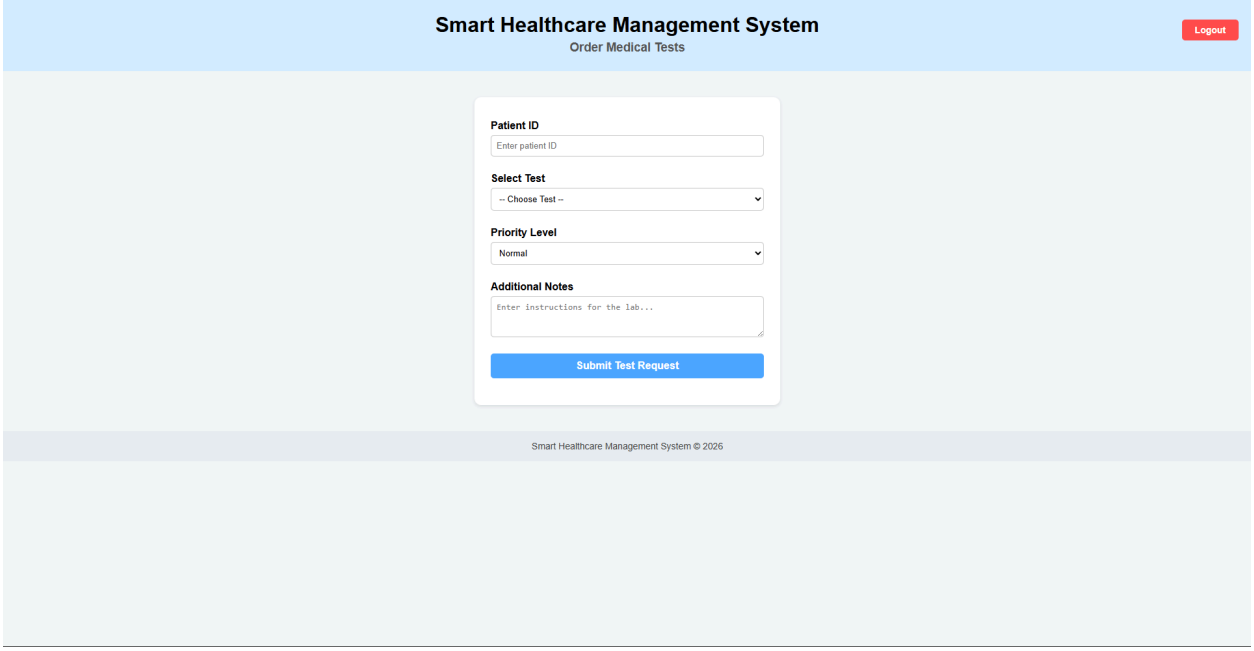


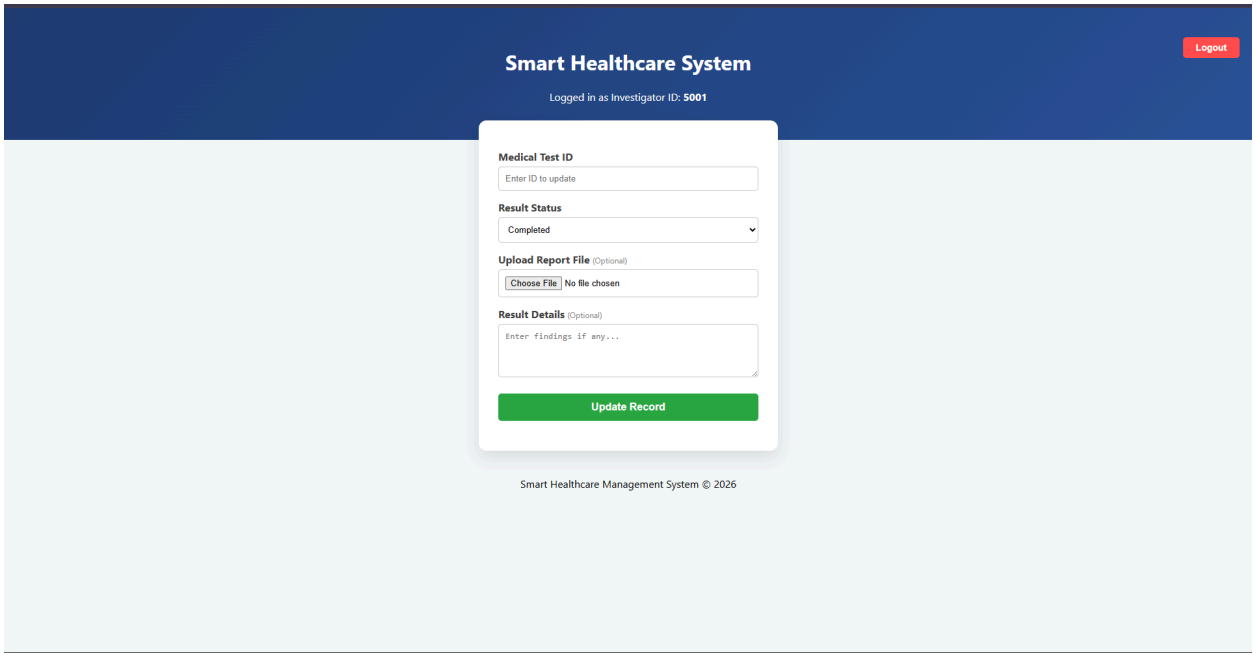


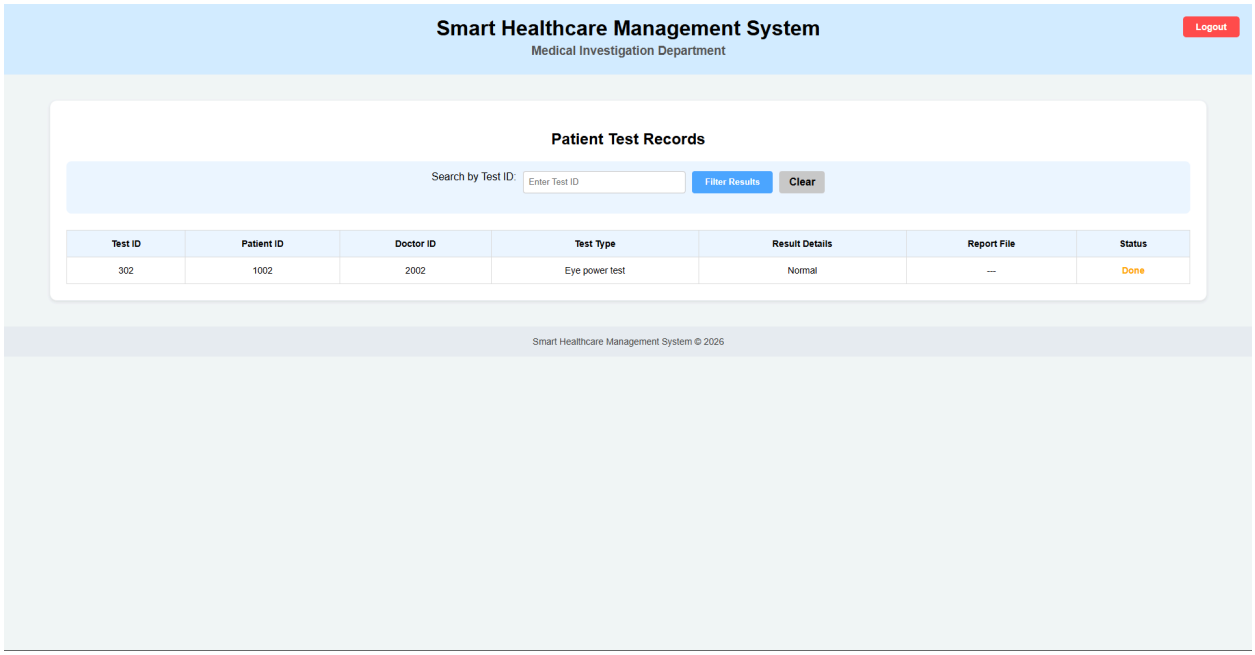
2.3 Webpage(s) for FEATURE 3

Webpage for FEATURE 3: Prescription & Investigation Management



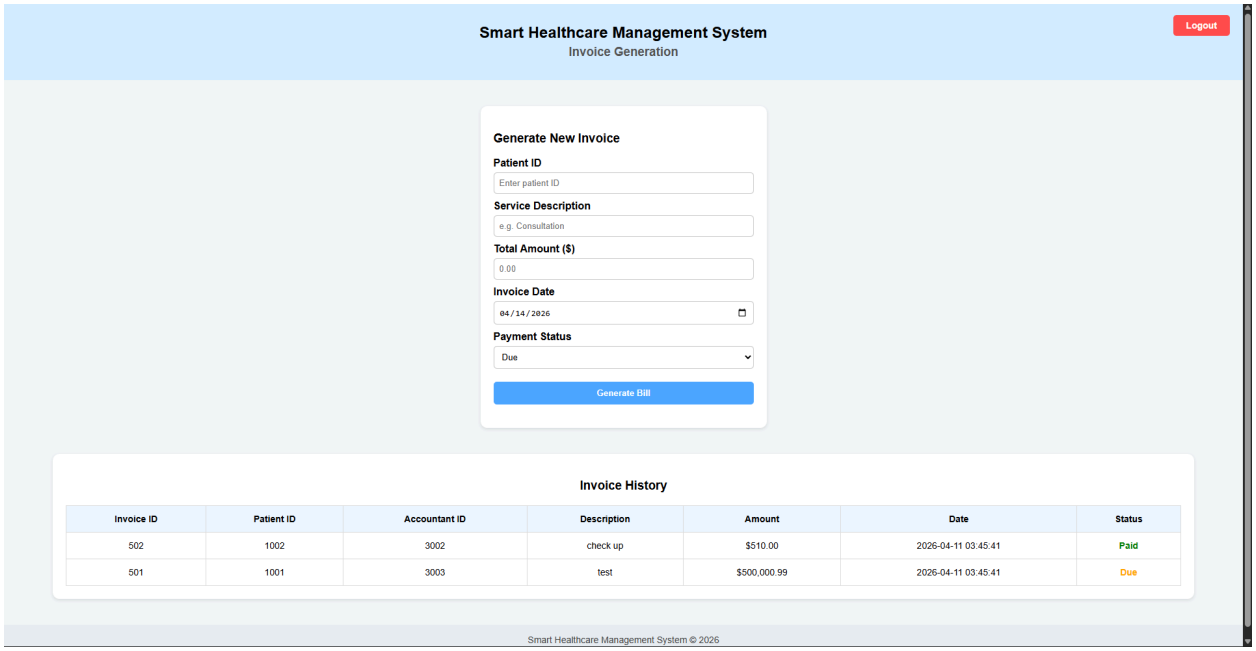

Smart Healthcare Management System
Order Medical Tests
Logout
Patient ID
Enter patient ID
Select Test
-- Choose Test --
Priority Level
Normal
Additional Notes
Enter instructions for the lab...
Submit Test Request
Smart Healthcare Management System © 2026

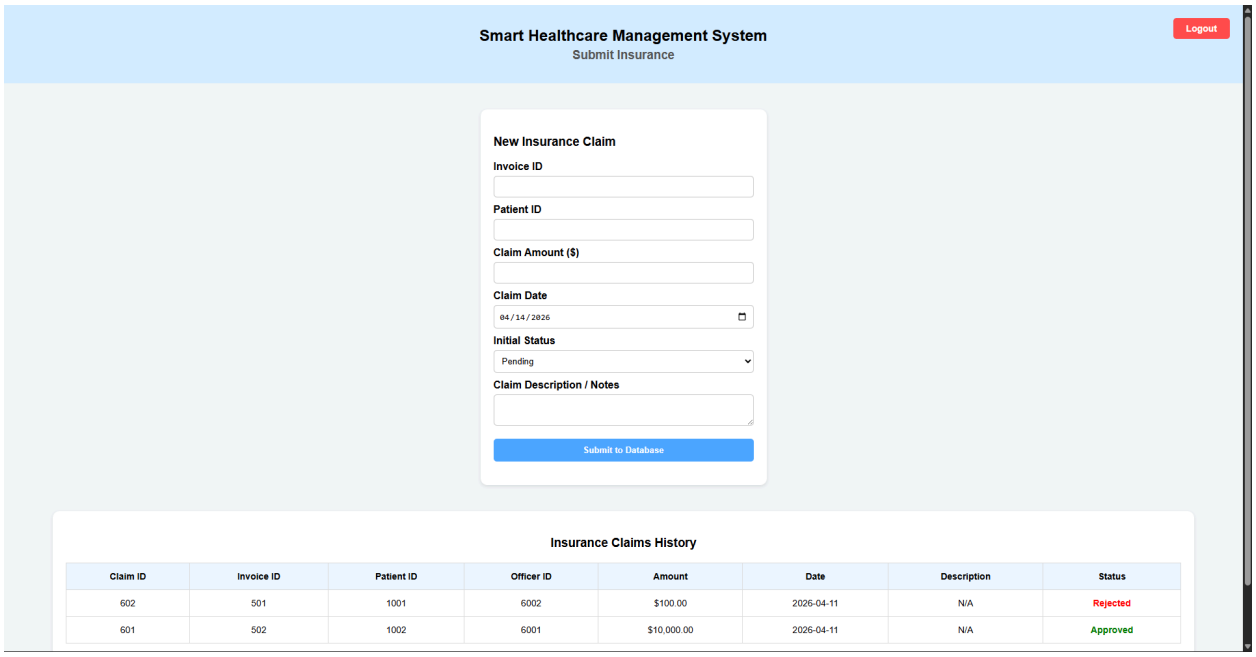

Logout
Smart Healthcare System
Logged in as Investigator ID: 5001
Medical Test ID
Enter ID to update
Result Status
Completed
Upload Report File (Optional)
Choose File No file chosen
Result Details (Optional)
Enter findings if any...
Update Record
Smart Healthcare Management System © 2026


Logout
Smart Healthcare Management System
Medical Investigation Department
Patient Test Records
Search by Test ID:
Enter Test ID
Filter Results
Clear
Test ID | Patient ID | Doctor ID | Test Type | Result Details | Report File | Status
302 | 1002 | 2002 | Eye power test | Normal | --- | Done
Smart Healthcare Management System © 2026

2.4 Webpage(s) for FEATURE 4

Webpage for FEATURE 4: Billing & Insurance Automation





Logout

Smart Healthcare Management System

Insurance Department

Submitted Insurance Claims

Filter Status: All Claims Filter Reset

Claim ID	Invoice ID	Patient Name	Phone Number	Claim Date	Claim Amount	Status
602	501	D Khan	1745453339	2026-04-11	$100.00	Rejected
601	502	M Rahman	1684628880	2026-04-11	$10,000.00	Approved

Smart Healthcare Management System © 2026

Logout

Smart Healthcare Management System

Insurance Department - Claims Management

Search by Claim ID... Search Reset

Update Claim

Claim ID (To Update)

e.g. 601

Update Status

Pending

Settlement Amount ($)

0.00

Update Description / Remarks

Enter notes here...

Update Record

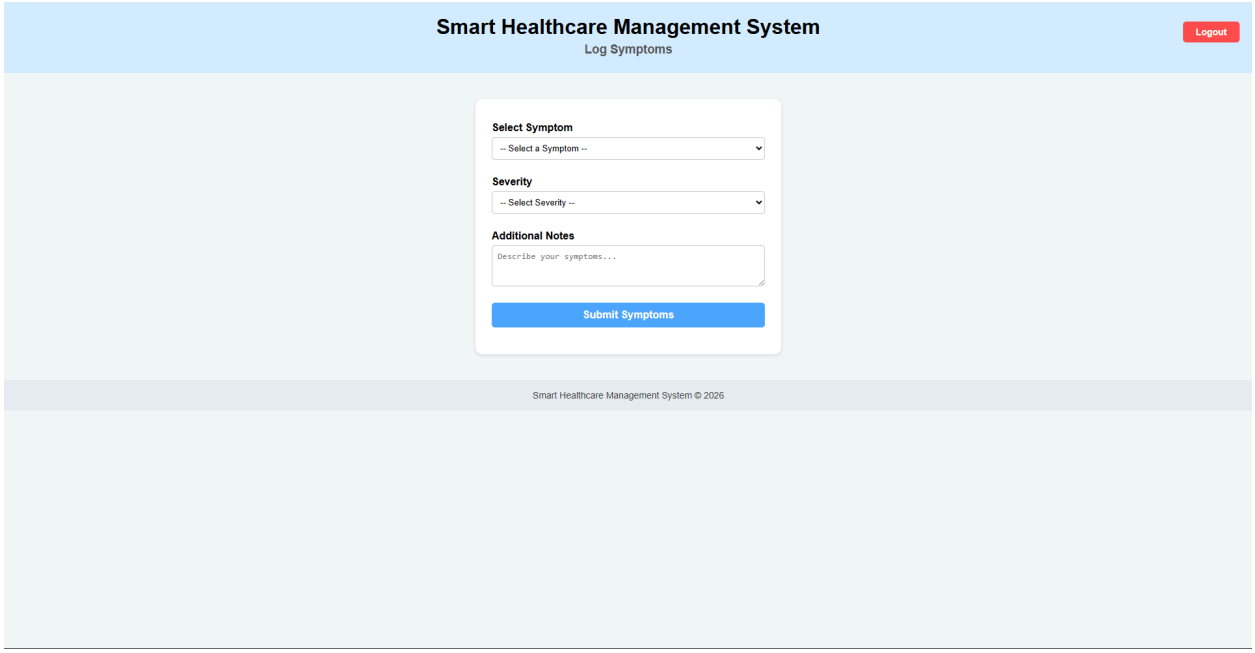
Live Claims Records

Claim ID	Patient ID	Amount	Date	Status
602	1001	$100.00	2026-04-11	Rejected
601	1002	$10,000.00	2026-04-11	Approved

Smart Healthcare Management System © 2026

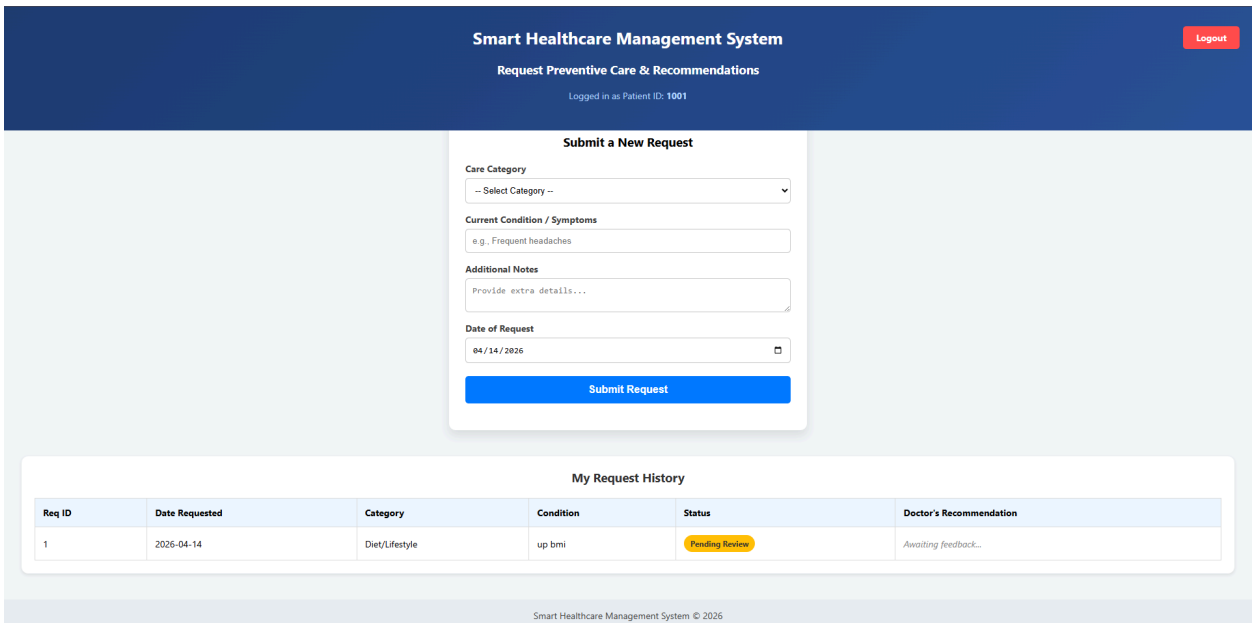
2.5 Webpage(s) for FEATURE 5

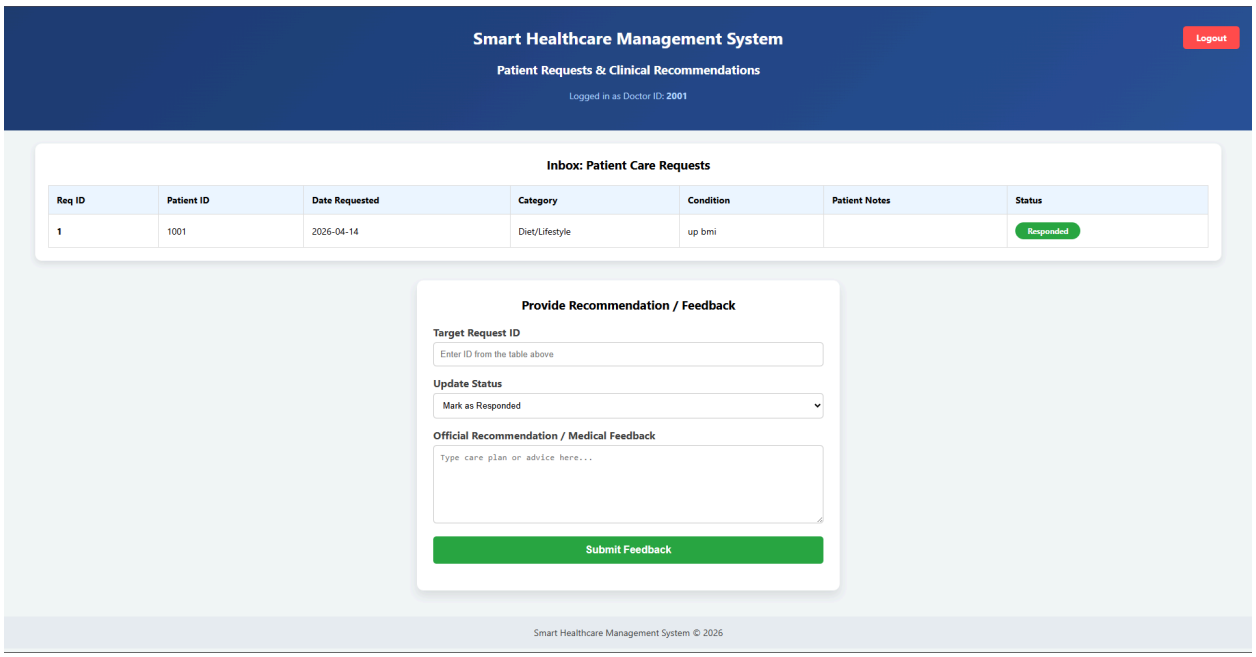
Webpage for FEATURE 5: Symptom & Disease



2.6 Webpage(s) for FEATURE 6

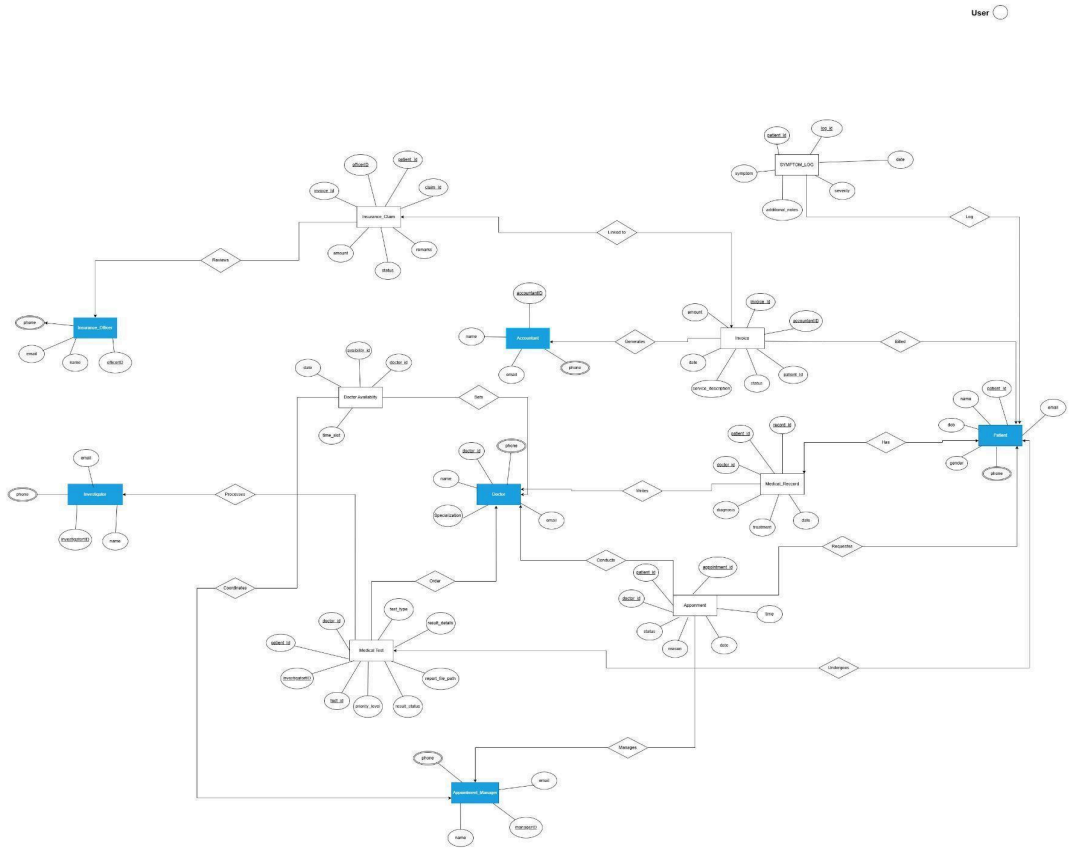
Webpage for FEATURE 6: Recommendations & Preventive Care





ENTITY RELATIONSHIP DIAGRAM (ERD)

3.1 Entity Relationship Diagram (ERD)



3.2 Explanation

1. Core Entities and Explanation

Patient

Stores patient information including personal details, contact information, and identification data. It acts as the central entity for appointments, records, billing, and insurance.

Doctor

Contains doctor details such as name, specialization, and contact information. Doctors are responsible for consultations, diagnoses, prescriptions, and test orders.

Appointment

Represents scheduled meetings between patients and doctors, including appointment date, time, reason, and status.

Medical Record

Stores consultation outcomes such as diagnosis, treatment plans, and visit history for each patient.

Medical Test & Investigator

Medical Test stores all ordered tests for patients, while Investigator manages the processing and uploading of test results.

Invoice & Insurance Claim

Invoice records billing information for hospital services, and Insurance Claim handles reimbursement requests processed by insurance officers.

Symptom Log

Allows patients to log symptoms before appointments to help doctors review conditions in advance.

2. Cardinality Summary

One Patient → Many Appointments

A patient can book multiple appointments over time.

One Doctor → Many Appointments

A doctor can conduct multiple appointments.

One Patient → Many Medical Records

Each patient may have several consultation records.

One Doctor → Many Medical Records

Doctors can create multiple records for different patients.

One Doctor → Many Medical Tests

A doctor may order multiple tests.

One Investigator → Many Medical Tests
An investigator can process multiple test reports.

One Patient → Many Invoices
A patient may receive multiple bills.

One Invoice → Many Insurance Claims
A single invoice may generate one or more insurance claims.

One Doctor → Many Availability Slots
Doctors can define multiple available schedules.

3. Operational Flow in the ERD

Appointment Scheduling

The workflow starts when a patient requests an appointment with a doctor based on available slots.

Consultation & Diagnosis

After the appointment, the doctor creates a medical record containing diagnosis and treatment details.

Test Processing

If required, the doctor orders medical tests, which are handled by investigators who upload results.

Symptom Logging

Patients can submit symptom logs before appointments for preliminary assessment.

Billing Process

Once treatment is completed, the accountant generates an invoice for services provided.

Insurance Handling

If insurance applies, an insurance claim is created and processed by the insurance officer.

DATABASE WITH TABLE (populate the table(s) with data)

4.1 Table for FEATURE 1

Table for FEATURE 1: Patient & Medical Record Management

This feature maintains comprehensive patient profiles and tracks their historical clinical diagnoses and treatment plans.

Table Name: patient

Patient_ID	Patient_Name	dob	Gender	Email_Address	Phone_Number
1001	D Khan	2000-04-15	Male	dk1001@private.com	1745453339
1002	M Rahman	2003-12-31	Female	mr1002@private.com	1684628880

Table Name: medical_record

record_id	report_ID	patient_id	doctor_id	diagnosis	treatment	date
406	303	1002	2002	Blood Test	Vitamin Supplement	2026-04-11
407	304	1001	2001	NULL	NULL	2026-04-14

4.2 Table for FEATURE 2

Table for FEATURE 2: Doctor & Appointment Management

This module manages doctor specializations and availability while coordinating the scheduling of patient visits through administrative managers.

Table Name: doctor

Doctor_ID	Doctor_Name	Specialization	Phone_Number	Email_Address	Status
2001	J Epstein	Surgical	1145453339	je2001@private.com	Free
2002	Walter White	Heart Surgeon	1484628880	ww2002@private.com	Busy

Table Name: appoinment

appointment_id	Doctor_ID	Patient_ID	status	reason	date	Manager_ID
101	2002	1002	Approved	fever	2026-04-10	4001
105	2002	1001	Pending	fever	2026-04-18	4001

4.3 Table for FEATURE 3

Table for FEATURE 3: Prescription & Investigation Management

This system handles the request and tracking of diagnostic medical tests, such as X-rays and blood tests, assigned to specific investigators.

Table Name: medical_record

record_id	report_ID	patient_id	doctor_id	diagnosis	treatment	date
406	303	1002	2002	Blood Test	Vitamin Supplement	2026-04-11
407	304	1001	2001	NULL	NULL	2026-04-14

Table Name: medicaltest

test_ID	Doctor_ID	Patient_ID	Test_Type	InvestigatorID	Priority _Level	Result_Status
301	2001	1001	X-Ray	5001	Urgent	Completed
303	2002	1002	Blood Test	5001	Urgent	In Progress

4.4 Table for FEATURE 4

Table for FEATURE 4: Billing & Insurance Automation

This feature automates the financial lifecycle by generating patient invoices and processing related insurance claims for approval or rejection.

Table Name: invoice

invoice_id	accountantID	patient_id	amount	status	date
501	3003	1001	500000.99	Due	2026-04-10
502	3002	1002	510.00	paid	2026-04-10

Table Name: insurance_claim

Claim_ID	Invoice_ID	Patient_ID	Amount	Officer_ID	Status	date
601	502	1002	1000.00	6001	Approved	2026-04-11
604	501	1001	500000.00	6001	Pending	2026-04-17

4.5 Table for FEATURE 5

Table for FEATURE 5: Symptom & Disease

This functionality allows patients to log specific health symptoms and their severity to assist in monitoring their ongoing medical condition.

Table Name: symptomlog

log_ID	Patient_ID	symptom	severity	date	additional_notes
202	1002	Losing Mind	Urgent	2026-04-15	Hallucinate Sometimes
205	1001	Fever	Severe	2026-04-14	Null

4.6 Table for FEATURE 6

Table for FEATURE 6: Recommendations & Preventive Care

This module facilitates patient inquiries regarding lifestyle or general health and stores official medical recommendations provided by doctors.

Table Name: care_requests

Req_ID	Patient_ID	Category	Patient_Condition	Status	Doctor_Recommendation
1	1001	Diet/Lifestyle	Up BMI	Responded	Eat Healthy avoid junk food
2	1001	General Health	Raising Weight	Pending Review	Null

GITHUB REPOSITORY & PROJECT DEMONSTRATION VIDEO

5.1 Github Repository

URL: Git hub: https://github.com/rashad7274/HealthCare_Management.git

5.2 Project Demonstration Video

URL: Group15_PS5_Demostration Video (UPDATED).mp4